

A median cap for gaps of a tiled integer packing

The inequality $\text{med}_i \leq \frac{8}{3} \text{sep}_i$, proved as far as packing goes, and the precise obstruction beyond

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Abstract

On a finite string of contiguous, pairwise disjoint blocks of positive integers, write n_i for the length of block i , c_i for its centre, gap_i for the one-sided nearest-centre distance, and sep_i for the cut-geometry lower bound $\min_j (n_i + n_j)/2$ on the two adjacent sides. Normalise by the radius-5 window median of neighbouring gaps. The inequality $\text{med}_i \leq \frac{8}{3} \text{sep}_i$ holds on every atom of a specific tiled packing coming from a folded discrete measure (0 violations among 32 728 atoms, the constant saturated). It is *not* a theorem of interval geometry: already the tiled composition $n = (1, 2, 8^{10})$ violates it, with ratio $16/3$. What packing *does* prove is a clean reduction: if the blocks tile an interval (no unoccupied integers between or inside blocks), then $\text{gap}_i = \text{sep}_i$ identically and the inequality becomes a statement about the length sequence (n_i) alone. That tiling identity holds with 0 exceptions on 134 windows (15 428 atoms) of the source construction. The remaining obstruction is named: the length sequence must forbid a C_2 jump (a $(1, 2)$ -pair of short blocks looking, inside the radius-5 window, at three or more large neighbour-gaps). Sign-run pairing of an arbitrary $\pm$ colouring of the tiled interval does *not* forbid C_2 . This note is a proof attempt with a named remainder, not a claim.

Status. Lemma 1 is proved (cut geometry). Lemma 2 is proved (tiling $\Rightarrow$ gap = sep). Lemma 3 is proved (an explicit tiled counterexample). Lemma 4 is proved (the saturating prefix, by direct computation). Proposition 1 is a finite census, not a cofinal law. The median-cap inequality itself remains open as a theorem; it is reduced to the named combinatorial invariant X_n of §6. Nothing in this document is evidence for or against the Riemann hypothesis.

1 The packing

Let $m \geq 2$. A *block packing* on $\mathbb{Z}$ is a sequence of nonempty finite intervals

$$I_i = \{L_i, L_i + 1, \dots, L_i + n_i - 1\} \subset \mathbb{Z}, \quad n_i \in \mathbb{N}_{\geq 1}, \quad i = 0, \dots, m - 1,$$

pairwise disjoint, ordered so that $L_0 < L_1 < \dots < L_{m-1}$. Write $R_i = L_i + n_i - 1$ for the right endpoint. The *unoccupied stretch* between adjacent blocks is the nonnegative integer

$$e_i = L_{i+1} - R_i - 1 \in \mathbb{N} \cup \{0\}.$$

The packing is *tiled* when $e_i = 0$ for every i (the union $\bigcup I_i$ is itself an interval). The *centre* of block i is the unweighted mean of its integers,

$$c_i = L_i + \frac{n_i - 1}{2} \in \frac{1}{2}\mathbb{Z}. \tag{1}$$

The adjacent *sides* and the one-sided *gaps* are

$$d_i = c_{i+1} - c_i, \quad \text{gap}_i = \begin{cases} d_0, & i = 0, \\ \min(d_{i-1}, d_i), & 1 \leq i \leq m-2, \\ d_{m-2}, & i = m-1. \end{cases} \quad (2)$$

The *separation* of an adjacent pair and the per-block separation are

$$s_i = \frac{n_i + n_{i+1}}{2}, \quad \text{sep}_i = \begin{cases} s_0, & i = 0, \\ \min(s_{i-1}, s_i), & 1 \leq i \leq m-2, \\ s_{m-2}, & i = m-1. \end{cases} \quad (3)$$

1.1 The radius-5 window median

Fix the window radius $W = 5$. For each index i let

$$N_i = \{ \text{gap}_k : |k - i| \leq 5, k \neq i, 0 \leq k < m \}$$

and write $K = |N_i|$. Sort N_i increasingly as $g_{(1)} \leq \dots \leq g_{(K)}$. The window median is the ordinary odd-count median, or the midpoint of the two central values when K is even:

$$\text{med}_i = \begin{cases} g_{(K/2+1/2)}, & K \text{ odd}, \\ \frac{1}{2}(g_{(K/2)} + g_{(K/2+1)}), & K \text{ even}. \end{cases} \quad (4)$$

In the interior of a long packing one has $K = 10$ and med_i is the mean of the 5th and 6th order statistics. At a left endpoint with $m \geq 6$ one has $K = 5$ and med_0 is the *third-smallest* of five neighbour-gaps. (The saturating atom of §4 is of this edge type.)

Definition 1 (MED-CAP). *A packing satisfies MED-CAP when*

$$\text{med}_i \leq \frac{8}{3} \text{sep}_i \quad \text{for every } i = 0, \dots, m-1. \quad (5)$$

The constant $8/3$ is not fitted. It is the reciprocal of the candidate floor $3/8$ for the normalised gaps $g_i = \text{gap}_i / \text{med}_i$: the chain $g_i \geq \text{sep}_i / \text{med}_i \geq 3/8$ closes pointwise if and only if both the separation lower bound of Lemma 1 and MED-CAP hold.

2 What packing proves

Lemma 1 (SEP-SATZ). *In every block packing, $d_i = e_i + s_i \geq s_i$, hence $\text{gap}_i \geq \text{sep}_i$ for every i . Equality holds in the side identity if and only if $e_i = 0$.*

Proof. From (1),

$$d_i = \left(L_{i+1} + \frac{n_{i+1}-1}{2} \right) - \left(L_i + \frac{n_i-1}{2} \right) = L_{i+1} - L_i + \frac{n_{i+1}-n_i}{2}.$$

Substitute $L_{i+1} = R_i + 1 + e_i = L_i + n_i + e_i$:

$$d_i = n_i + e_i + \frac{n_{i+1}-n_i}{2} = e_i + \frac{n_i+n_{i+1}}{2} = e_i + s_i.$$

The gap is a minimum of one or two sides, the separation is the same minimum of the corresponding s -values, and $e \geq 0$ gives the inequality. $\square$

Equivalently: a mean of n distinct integers all lying on one side of a cut sits at least $(n-1)/2$ away from the cut when the integers are packed against it, and cannot sit closer. Adjacent blocks, being disjoint and ordered, are separated by a cut.

Lemma 2 (tiling implies gap = sep). *If the packing is tiled ($e_i = 0$ for all i), then $d_i = s_i$ for every adjacent pair, and therefore $\text{gap}_i = \text{sep}_i$ for every block. In that case MED-CAP is equivalent to*

$$\text{med}_i \leq \frac{8}{3} \text{gap}_i \quad \text{for every } i,$$

a statement that refers only to the integer sequence $(n_i)_{i=0}^{m-1}$.

Proof. Lemma 1 with $e_i = 0$. □

Thus any proof of MED-CAP from packing axioms alone must either use a positive lower bound on the unoccupied stretches (which *raises* sides and cannot help an upper bound on the median) or constrain the length sequence (n_i) . Empty stretches can only make MED-CAP harder: they inflate neighbouring gaps while leaving sep_i unchanged. The useful packing axiom, if there is one, is tiling plus regularity of (n_i) .

3 What packing does not prove

Lemma 3 (C_2 counterexample). *The tiled packing of $m = 12$ blocks with lengths*

$$n = (1, 2, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8)$$

(so $e_i = 0$ for all i) satisfies SEP-SATZ with equality on every side, and violates MED-CAP at the two leftmost blocks, with ratio $\text{med}_i/\text{sep}_i = 16/3 > 8/3$.

Proof. The sides are $d_0 = (1+2)/2 = 3/2$ and $d_i = 8$ for $i \geq 2$, with $d_1 = (2+8)/2 = 5$. The gaps are therefore $\text{gap}_0 = \text{gap}_1 = 3/2$ and $\text{gap}_i = 8$ for $i \geq 2$ except $\text{gap}_2 = \min(5, 8) = 5$. At $i = 0$ the radius-5 window has five neighbours, with sorted gaps $(3/2, 5, 8, 8, 8)$; the third-smallest is 8. The separation is $\text{sep}_0 = 3/2$, and $8/(3/2) = 16/3 > 8/3$. The same median 8 appears at $i = 1$ against the same separation. □

A still shorter tiled witness is $n = (1, 2, 6, 6, 6)$: the five neighbour-gaps at the left edge are $(3/2, 4, 6, 6, 6)$, the third-smallest is 6, and $6/(3/2) = 4 > 8/3$. Random tiled compositions with $n_i \leq 9$ and no Lipschitz constraint violate MED-CAP in roughly one third of draws; imposing the observed jump bound $|n_{i+1} - n_i| \leq 12$ does not remove the violators. Packings with a single large unoccupied stretch (the three singletons at $\{0\}$, $\{1\}$, $\{10\}$) likewise violate, but those are excluded the moment one assumes tiling.

Consequently MED-CAP is *not* a theorem of disjoint contiguous supports plus SEP-SATZ, and it is *not* a theorem of tiled compositions of an interval. Any proof must use a further constraint on (n_i) .

4 The equality case

The constant $8/3$ is sharp. Direct computation:

Lemma 4 (saturating prefix). *The tiled packing whose first six lengths are*

$$n = (1, 2, 6, 5, 4, 4)$$

has, at the left endpoint, $\text{gap}_0 = \text{sep}_0 = 3/2$ and $\text{med}_0 = 4$, hence

$$\frac{\text{med}_0}{\text{sep}_0} = \frac{4}{3/2} = \frac{8}{3}.$$

The same median and separation occur at $i = 1$ (six neighbours, even window: mean of the two central values).

Proof. Sides: $d_0 = 3/2$, $d_1 = (2 + 6)/2 = 4$, $d_2 = (6 + 5)/2 = 11/2$, $d_3 = (5 + 4)/2 = 9/2$, $d_4 = 4$. Gaps: $\text{gap}_0 = 3/2$, $\text{gap}_1 = \min(3/2, 4) = 3/2$, $\text{gap}_2 = \min(4, 11/2) = 4$, $\text{gap}_3 = \min(11/2, 9/2) = 9/2$, $\text{gap}_4 = \min(9/2, 4) = 4$, $\text{gap}_5 = 4$. The five neighbours of $i = 0$ sort as $(3/2, 4, 4, 4, 9/2)$; the third is 4. The six neighbours of $i = 1$ sort as $(3/2, 4, 4, 4, 4, 9/2)$; the mean of the 3rd and 4th is 4. $\square$

This prefix is realised by the folded packing of the Dirichlet character χ_4 at the admissible window of index 53 ($N = 606$, $m = 113$ blocks, 496 occupied grid integers, tiling exact, two atoms with $g_i = 3/8$). The bulk mode of that packing is $n_i = 4$ (and $n = 4$ is the mode on every tested world, §5). The arithmetic

$$\frac{8}{3} = \frac{4}{3/2}$$

is exactly bulk-mode pair-separation over the edge $(1, 2)$ -separation. Replacing the tail $(5, 4, 4)$ by $(6, 6, 6)$ produces the short C_2 witness of §3: the third-smallest neighbour-gap jumps from 4 to 6. Saturation is therefore a *one-sided* edge phenomenon — a short $(1, 2)$ pair, a single overshoot to $n = 6$, and an immediate return to the bulk mode 4.

5 Tiling on the source packing

The packings that motivated the inequality are the level-2 block decompositions of four families of folded discrete measures (a prime-power comb, a second oversampling, and two Dirichlet matchings), after an edge mask, on a finite list of windows. Block supports are the distinct integer images of the atoms under the grid map $t = \arccos(x) \cdot (4N - 2)/(2\pi)$, grouped by paired sign runs in x -order. The map is injective on the bulk up to the quantisation bound 10^{-12} ; centres, sides, gaps and medians are therefore exact rationals.

Proposition 1 (tiling census). *On 134 windows comprising 15 428 blocks — every admissible window of both Dirichlet matchings $(42 + 42)$, the 42-rung core ladder of the prime-power comb, and the eight sealed anchors of the second family — one has $e_i = 0$ for every adjacent pair (15 294 sides) and every block hull is solid (no unoccupied integer inside any I_i). Consequently $\text{gap}_i = \text{sep}_i$ identically on those 15 428 atoms, and MED-CAP holds with 0 violations, worst ratio exactly $8/3$ at the two atoms of Lemma 4.*

The same tiling identity is the reason the original 32 728-atom census (all 181 windows of the four families, including 47 extension rungs of the prime-power comb not re-enumerated here) recorded 0 MED-CAP violations: under tiling the inequality depends only on (n_i) , and that sequence, on the source, avoids C_2 (Proposition 2). Tiling was not used in the original census; it is the packing fact the census was pointing at.

A structural reason for tiling, not used as a hypothesis below, is available at the level of the grid map: after the edge mask the surviving atoms occupy a convex x -interval, $\arccos$ is strictly monotone, and empirically every integer of the resulting t -interval is attained. A vanishing theorem for the underlying trigonometric symbol on that bulk interval would promote Proposition 1 from census to construction algebra. That promotion is not carried out here.

6 The remaining invariant

Write X_{tile} for the assertion that the packing is tiled and hull-solid. By Lemma 2,

$$X_{\text{tile}} + \text{MED-CAP} \iff X_{\text{tile}} + X_n,$$

where X_n is the assertion that the length sequence (n_i) is $(8/3)$ -median-regular in the sense of (5) with $\text{sep}_i = \text{gap}_i$. Lemma 3 says X_n is not true of every composition of an interval.

Two sufficient conditions that *would* imply X_n both fail on the source, or fail to be sufficient:

- *Local oscillation of gaps.* If $\max / \min$ of the gaps in every radius-5 window were at most $8/3$, MED-CAP would follow at once (the window median cannot exceed the window maximum). At the saturating prefix of Lemma 4 one has $\max / \min = (9/2)/(3/2) = 3 > 8/3$. The bound on the *maximum* is too crude; MED-CAP bounds an order statistic.
- *Interior restriction $n_i \in \{3, 4, 5\}$.* Then every interior side lies in $[3, 5]$ and every interior ratio is at most $5/3 < 8/3$. On the source this fails at several hundred interior blocks per family (mode $n = 4$ is only about 58 percent; $n = 2$ occurs in the interior, especially on the second family). Those interior $n = 2$ blocks sit next to $n = 4$ neighbours, so $\text{sep} = 3$ and the budget $(8/3) \cdot 3 = 8$ is generous — they are not C_2 .
- *Sign-run pairing.* On the source, blocks are paired consecutive sign-runs of a $\pm$ colouring of the tiled interval (an odd tail is a block by itself). That algebra yields $n = 1$ only as an unpaired singleton tail and $n = 2$ as a pair of singleton runs, which is compatible with the saturating prefix. It does *not* forbid C_2 : the colouring “three singleton runs, then a run of length 8” is a legal sign sequence and realises $n = (1, 2, 8, \dots)$ at the odd-tail edge.

Proposition 2 (no C_2 on the source). *Among the 15 428 atoms of Proposition 1 there are 114 blocks with $n_i \leq 2$ that see some $n_j \geq 7$ inside their radius-5 window. None of them violates MED-CAP (worst such ratio $12/5 < 8/3$, at the left edge of the χ_3 window of index 20, prefix $(1, 4, 8, 4, 8, 4)$, separation $5/2$). The unique atoms that saturate $8/3$ are the two of Lemma 4. Adjacent $(n, n') = (1, 2)$ with a later window value ≥ 8 does not occur.*

The named remainder is therefore:

Invariant X_n . The length sequence of the tiled packing does not contain, at any small-separation locus ($\text{sep}_i = 3/2$, i.e. an adjacent $(1, 2)$ or $(2, 1)$ pair, or $\text{sep}_i = 1$ from two adjacent singletons), a radius-5 window whose third-smallest neighbour-gap (edge) or mean of the 5th and 6th (interior) exceeds $(8/3) \text{sep}_i$. Equivalently: after an edge $(1, 2)$ pair, an overshoot to $n = 6$ is permitted only if the sequence returns to the bulk mode 4 fast enough that at most two of the five neighbour-gaps exceed 4; an overshoot to $n \geq 7$ at that locus is forbidden.

X_n is a property of the *sign field* of the folded measure (or of whatever mechanism produces the observed n -histogram, mode 4, geometric-like run lengths), not of interval packing. A proof of X_n is a proof of MED-CAP on every tiled packing that the construction produces. A counterexample to X_n on a source window would be a MED-CAP violation, and none is known.

7 T1 preview

Assume MED-CAP. Then $g_i = \text{gap}_i / \text{med}_i \geq 3/8$ pointwise, so the dyadic packing count $S_r = \sum_{g_i \leq 2^{-r}} q_i$ vanishes for every $r \geq 2$ (since $2^{-2} < 3/8$), and the cubic mass of the block profile satisfies

$$M_3 \leq \left(\frac{8}{3}\right)^2 C_K^2 \frac{(\log m)^2}{m^2}$$

once the localisation bound $(mq_i / \log m) g_i \leq C_K$ is granted. Two logarithms and the second Carleson constant drop. What does *not* drop is C_K itself, and the present attempt explains why a packing proof of MED-CAP would not have produced it.

MED-CAP, even after the tiling reduction, is an order-statistic bound on the *length sequence* of blocks, independent of the block masses q_i . The T1 bound is a statement about those masses relative to the same gaps. A purely geometric MED-CAP (which this note refutes) would have suggested that T1 might also be local packing; the obstruction X_n being a sign-field constraint says the opposite. The masses q_i see the same sign field through the defect measure, but they see it as an amplitude, not as a run length. Proving X_n does not create a family-uniform C_K , and the r358 record that the per-family freeze maximum fails on the test set (ceiling 23.70 against freeze 4.91) stands unchanged. The single remaining quantitative object of the terminal lane, after MED-CAP, is that T1 constant rule — to be addressed on its own terms, not as a corollary of the median cap.

8 Machine checks

Every numbered lemma is re-proved by exact rational arithmetic in the companion script `rh/problem/verify_medcap_steps.py`. The script also rebuilds the saturating prefix from the source window χ_4 at index 53 and re-gates tiling plus MED-CAP on the four shallow windows of index 9. The 134-window tiling census of Propositions 1 and 2 is the record of a one-shot enumeration against the same Fractions ledger (not a sealed probe). Exit line: MEDCAP STEPS VERIFIED.

Claim boundary

Finite identities and a named combinatorial remainder on a finite family of explicit packings. No statement about zeros of $\zeta(s)$, in either direction. No RH claim.

References

- [1] S. Hamann, *A strict polynomial subordination inequality for two explicit prime-generated discrete measures*, standalone note, August 2026, `rh/problem/1star_problem.tex`.